

# Context-Aware Bias Detection and Controlled Text Rewriting for Fairer News Media

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## Abstract

Media bias is a pervasive issue that influences public perception and political discourse. Existing AI-based bias detection models primarily focus on classification but lack effective solutions for bias mitigation. This study explores transformer-based models to enhance the detection of biased content while ensuring transparency and journalistic integrity. The initial phases involved training models on large-scale datasets; however, computational constraints necessitated a transition to a more manageable dataset. We present our findings from training a BERT-based bias detection model, evaluating its performance, and outlining the next steps toward bias mitigation.

## 1 Introduction

The spread of biased news articles across digital platforms amplifies ideological slants and reinforces existing beliefs (Goel et al., 2023). This study addresses the issue by implementing a two-stage AI-driven system: bias detection and bias mitigation through controlled text generation. The first phase involves developing a robust classification model to identify both explicit and implicit bias. Due to computational constraints with large-scale datasets, we transitioned to a more manageable and structured dataset, enabling successful model training and evaluation. These initial results informed refinements to the detection process ahead of the mitigation phase.

Our research is guided by the ethical question: *Can AI reliably detect and mitigate bias in news media without compromising factual accuracy and journalistic expression?* Key challenges include defining an objective and generalizable measure of bias across media outlets and political contexts, and ensuring that

mitigation does not introduce new biases or distort factual content. Additionally, the ethical implications of AI-driven rewriting must be considered, particularly regarding its influence on editorial voice and intent. These concerns underscore the complexity of building AI systems that can detect, explain, and reduce bias while maintaining the neutrality and credibility of news reporting.

## 2 Related Work

Bias detection in news media has been widely studied using NLP, particularly with transformer models. (Khouja, 2020) used BERT embeddings and metadata for stance detection, improving accuracy. Lexicon-based approaches (Kulkarni et al., 2018) analyzed linguistic framing via word embeddings and sentiment scores but faced challenges with context. Graph-based methods like (Li and Goldwasser, 2019) modeled entity relationships to enhance political bias classification, while (Gentzkow and Shapiro, 2010) measured media slant through word frequency analysis. Model explainability enhances trust in bias detection systems. LIME (Ribeiro et al., 2016) and SHAP (Lundberg and Lee, 2017) help interpret model outputs by identifying influential features. Attention-based methods (Robertson et al., 2018) have also improved transparency in analyzing framing bias. Bias mitigation through text rewriting is emerging. (Kim et al., 2023) employed BART for controlled rewriting to reduce bias while preserving facts. Style transfer techniques (Pryzant et al., 2020), unsupervised text transformation (Lample et al., 2018), and reinforcement learning approaches

have furthered bias reduction by optimizing neutrality. Despite advances in detection, effective mitigation that preserves journalistic integrity remains limited. Our work addresses this by combining explainable detection with automated rewriting to support fair and factual media content.

### 3 Proposed Methodology

Our methodology comprises three key stages aimed at reducing media bias while preserving factual accuracy. First, bias detection is performed using a fine-tuned BERT model to classify articles as left, center, or right based on their title and content. Next, model interpretability is addressed using SHAP and LIME to highlight influential linguistic features, ensuring transparency in predictions. Finally, bias mitigation leverages a T5-based text generation model to rewrite biased content into neutral versions, evaluated using similarity metrics and reclassification checks to ensure semantic integrity and reduced bias.

#### 3.1 Dataset

Two datasets were used in our project.

1. Article-Bias-Prediction Dataset<sup>1</sup>: This dataset contains 37,554 politically labeled news articles (left, center, right) with titles and content concatenated as input. It was preprocessed via BERT tokenization, missing value removal, and stratified splitting, and used to train our BERT-based classifier.

2. Bias-Detection-Combined Dataset<sup>2</sup>: This domain-agnostic dataset of 16,000 biased-neutral sentence pairs was used to train the T5 model for rewriting biased text while preserving meaning and factual accuracy.

#### 3.2 Bias Detection

To detect political bias in news articles, we fine-tuned a pre-trained BERT-base-uncased transformer model using the Article-Bias-Prediction dataset. The dataset was stratified and split into training (80%), validation (10%),

and test (10%) sets to maintain class balance. The model was trained using a cross-entropy loss function with the AdamW optimizer.

An initial baseline model was trained for five epochs with a batch size of 8, achieving an accuracy of **81%**. While it performed reasonably well for left- and right-biased articles, it struggled with center-biased content. The confusion matrix for the base BERT model is shown in Appendix Figure 3. It illustrates strong classification performance for left and right classes, with some confusion in the center class. The fine-tuned model, trained for four epochs with a learning rate of  $5e^{-5}$  and a batch size of 16, achieved a significantly improved accuracy of **92.78%** and an F1-score of **0.9278**. Evaluation was conducted after each epoch, with the best checkpoint selected based on F1 score.

This 11.78 percentage point improvement was driven by several key changes: hyperparameter optimization, concatenation of titles and content for richer contextual input, stratified sampling, and checkpointing based on F1 score.

The performance comparison between the initial and fine-tuned models is summarized in Table 1.

| Metric    | Base Model | Fine-Tuned Model |
|-----------|------------|------------------|
| Accuracy  | 0.8100     | <b>0.9278</b>    |
| F1 Score  | 0.8100     | <b>0.9278</b>    |
| Precision | 0.81       | <b>0.9278</b>    |
| Recall    | 0.81       | <b>0.9278</b>    |

Table 1: Performance Comparison: Initial vs. Fine-Tuned BERT Model

#### 3.3 SHAP and LIME Explainability

To enhance interpretability, we applied SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) to the fine-tuned BERT model. These methods identified which tokens most influenced the model’s bias classifications (left, center, right).

For LIME, a representative subset of test articles was perturbed to create linguistic variations, and a local surrogate model was trained

<sup>1</sup><https://github.com/ramybaly/Article-Bias-Prediction>

<sup>2</sup><https://huggingface.co/datasets/pranjali97/Bias-detection-combined>

to estimate token importance. Visual outputs included bar charts and highlighted text, as shown in Figure 4 and Figure 1.

SHAP, implemented via a transformer-compatible pipeline, provided global token attributions by comparing each token’s influence to expected baseline values. SHAP force plots visualized these contributions.

Both methods revealed that the model frequently relied on ideologically meaningful terms like “freedom,” “rule of law,” and “disruptions,” supporting that predictions were driven by semantically relevant cues rather than arbitrary patterns.

### 3.4 Bias Mitigation

The final stage of our methodology focuses on reducing bias in news content through controlled text rewriting using the T5 (Text-to-Text Transfer Transformer) architecture. We fine-tuned both T5-Base and T5-Large variants on the Bias-Detection-Combined dataset, which includes over 16,000 domain-agnostic, non-political sentence pairs of biased and neutral text. These examples provided general rewriting patterns for reducing ideological framing while preserving meaning.

T5-Base was chosen for its lower computational cost, while T5-Large—with more parameters—produced more fluent and coherent rewrites, especially on complex sentences. Training involved minimizing cross-entropy loss between biased inputs (prefixed with a task prompt) and their corresponding neutral targets.

We evaluated both models using ROUGE scores (ROUGE-1, ROUGE-2, ROUGE-L) and a content preservation score. Results in Table 2 show that T5-Base outperformed T5-Large across all metrics, despite the latter’s fluency advantage. Visual comparisons of these trends are shown in Figure 2 and Figure 5 from Appendix, highlighting the trade-off between fluency and factual retention.

### 3.5 Experimentation

The project initially aimed to use the full News-Media-Bias dataset<sup>3</sup> containing over 3.7 million entries. However, training transformer models at that scale proved infeasible due to computational constraints. This led to a revised approach involving more efficient model choices and reduced dataset size.

We adopted the Article-Bias-Prediction dataset (37.5K articles) for the bias detection task and used the BERT-base-uncased model due to its efficiency and lower resource demands. Preprocessing steps included combining article title and content, stratified splitting, and label encoding.

For bias detection, we experimented with RoBERTa, but it exhibited unstable convergence due to tokenizer mismatches and high resource consumption. The BERT-based model, in contrast, was successfully fine-tuned and achieved a substantial accuracy gain.

Bias mitigation experiments utilized the Bias-Detection-Combined dataset, comprising over 16,000 biased-neutral sentence pairs. We fine-tuned both T5-Base and T5-Large models. Rewritten outputs were evaluated using ROUGE and content preservation metrics.

## 4 Results and Analysis

The baseline BERT model achieved an accuracy of 81%, revealing difficulty in distinguishing subtle political bias—particularly in center-leaning articles, which were often misclassified as left or right. This limitation reflected challenges in recognizing neutral language that lacks strong ideological markers. The fine-tuned model, by contrast, achieved 92.78% accuracy, with marked improvements across all metrics. It showed better recognition of ideological cues and more consistent classification for polarized classes, though confusion in the center class persisted. This highlights the ongoing challenge of modeling political neutrality.

Explainability tools SHAP and LIME revealed that the model relied on semantically

<sup>3</sup><https://huggingface.co/datasets/newsmediabias/news-bias-full-data>

Now More than Ever, Free-Speech Dogma Is Worth Defending [SEP] Former Charlottesville, Va., mayor Michael Signer argues that we must restrict speech to maintain the rule of law. He couldn't be more wrong. "Two things form the bedrock of any open society," Salman Rushdie once noted, "freedom of expression and rule of law. If you don't have those things, you don't have a free country." Well, in a recent Wall Street Journal op-ed, "How Free Speech Dogma Failed Us in Charlottesville," Michael Signer, the former mayor of Charlottesville, makes the argument that restricting speech is necessary for the rule of law. The first problem with Signer's case is the premise itself. Sorry, but we have no uniquely pressing need to "keep pace" with violent or threatening "political disruptions." Americans live in an era of relatively little political violence. A person doesn't even have to go back to the brutality of the 1860s and 1870s to understand this; they can just look back at the 1960s and 1970s, or maybe even the 1990s. When a few hundred Nazis, in a nation of 350 million, get together and march down Main Street, that isn't a particularly compelling reason to rethink our rights. In 1939, well after Hitler's tyrannical intentions were known, 20,000 Nazi sympathizers filled up Madison Square Garden. There will always be extremists in America. Which is one reason why we must always do our best to safeguard natural rights. With white nationalists chanting "Jews will not replace us" in the streets of Charlottesville and far-left demonstrators

Figure 1: Highlighted Tokens Contributing to Model Prediction

| Model    | ROUGE-1 | ROUGE-2 | ROUGE-L | Content Preservation |
|----------|---------|---------|---------|----------------------|
| T5-Base  | 0.6053  | 0.5687  | 0.6022  | 0.6022               |
| T5-Large | 0.4171  | 0.3645  | 0.4103  | 0.4103               |

Table 2: Performance Metrics for Bias Mitigation Models

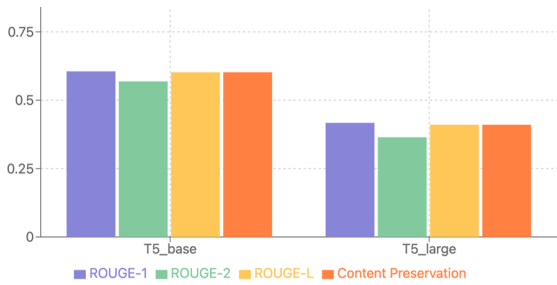


Figure 2: Model Performance Comparison — Bar Chart for T5-Base vs. T5-Large. T5-Base achieves higher ROUGE and content preservation scores, while T5-Large lags despite generating more fluent text.

meaningful and ideologically charged terms like “freedom,” “rule of law,” and “disruptions.” These consistently influenced predictions for left and right articles. In contrast, center-leaning predictions showed less consistent token attribution, reinforcing the ambiguity and modeling difficulty of centrist content. Overall, the explainability analysis validated the model’s internal logic for polarized texts but exposed interpretability gaps for neutral cases.

For bias mitigation, both T5-Base and T5-Large were fine-tuned to rewrite biased content into neutral versions. T5-Large produced more fluent and coherent rewrites, especially for complex sentences, while T5-Base better

preserved factual content—reflected in higher ROUGE and content preservation scores.

To illustrate, consider the sentence: “The liberal agenda continues to undermine our national values.” The fine-tuned BERT model classified it as right-biased. T5-Base rewrote it as “Current policies are changing traditional values,” which was still classified as right. T5-Large softened it to “Recent decisions have influenced national values,” which was reclassified as center-biased. This suggests T5-Large was marginally better at reducing ideological tone, though results were inconsistent.

We initially considered using the fine-tuned BERT classifier to reclassify T5-generated outputs as a proxy for mitigation effectiveness. However, this proved uninformative: most rewrites retained the original bias label, particularly those from T5-Base. This indicated that ideological framing was not sufficiently altered to affect the classifier’s output. Therefore, we excluded this method from our final evaluation due to its inability to capture subtle shifts in framing, especially given the models were trained on domain-agnostic data.

These findings suggest that while transformer-based rewriting shows promise for bias mitigation, its effectiveness is limited without domain-specific training. Future

work should explore politically annotated datasets and integrate human feedback for more accurate evaluation.

The full implementation and resources are available in the project’s GitHub repository<sup>4</sup>.

## 5 Future Work

Future work will focus on improving bias mitigation and enhancing system robustness. A key direction is fine-tuning the T5 model on politically annotated datasets, as results indicate that general, non-political sentence pairs are insufficient for nuanced debiasing. Human evaluation will also be critical to assess neutrality, fluency, and factual consistency beyond automated metrics. Further research may incorporate discourse-level features to detect subtler biases and include adversarial testing to evaluate model reliability. Exploring larger language models could improve fluency and contextual preservation in rewritten text.

## 6 Conclusion

This study proposed an end-to-end framework for detecting and mitigating political bias in news articles using transformer-based models. The fine-tuned BERT classifier achieved high accuracy and interpretability through SHAP and LIME, confirming its reliance on ideologically meaningful cues. While the T5 models showed initial promise for automated rewriting, their effectiveness was limited by non-political training data, resulting in outputs that often lacked fluency or retained bias. The results highlight the value of combining explainable detection with controlled rewriting and underscore the need for domain-specific enhancements to build more reliable and fair AI systems for media applications.

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<sup>4</sup>[https://github.com/Jayasri2021/EAI\\_News\\_Media\\_Bias](https://github.com/Jayasri2021/EAI_News_Media_Bias)

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## Appendix

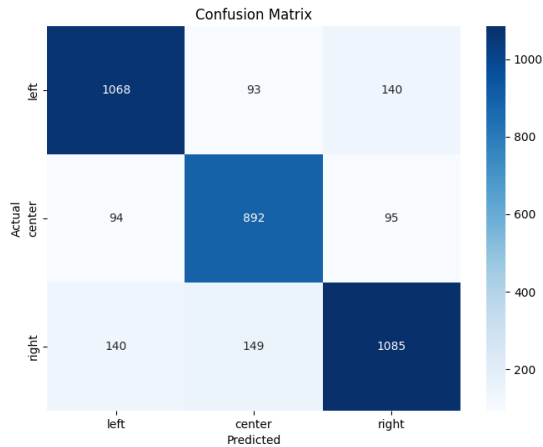


Figure 3: Confusion Matrix of Fine-Tuned BERT Model. The model accurately classifies left and right articles but shows confusion in identifying center-biased content.

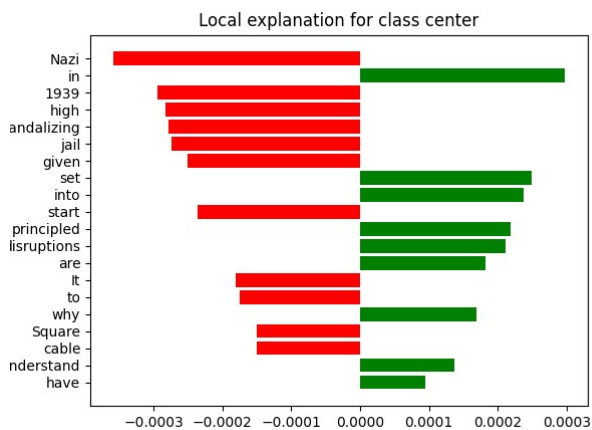


Figure 4: LIME Explanation for a Right-Leaning Prediction

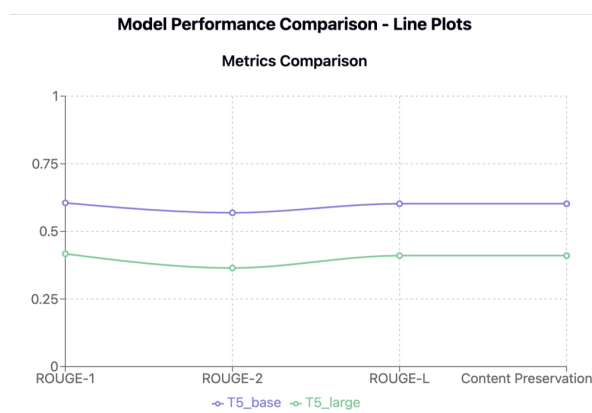


Figure 5: Model Performance Comparison — Line Plot for T5-Base vs. T5-Large. A visual comparison of each model's metric trends across ROUGE-1, ROUGE-2, ROUGE-L, and content preservation.